

Isometric classification

2D Hyperquadric: $C: g_{11}x^2 + g_{12}xy + g_{22}y^2 + b_1x + b_2y + c = 0$.

↓ rotation + translation

$$C: \lambda_1 x^2 + \lambda_2 y^2 = K, \quad \lambda_1 > 0.$$

I: $\lambda_2 > 0, K = 0$

$\Rightarrow (0,0)$ unique sol. $\Rightarrow$ The curve degenerates to a point = origin.

II: $\lambda_2 > 0, K < 0$

$\Rightarrow \nexists$ real sol.

III: $\lambda_2 > 0, K > 0$

$$\Rightarrow | : K \Rightarrow \frac{x^2}{\frac{K}{\lambda_1}} + \frac{y^2}{\frac{K}{\lambda_2}} = 1 \Rightarrow \text{Ellipse}$$

check can. form: $\frac{K}{\lambda_1} > \frac{K}{\lambda_2} \rightarrow \text{yes}$
 $\rightarrow \text{no} \rightarrow \text{rotate } 90^\circ \Rightarrow \text{DIRECT ISOMETRY.}$

IV: $\lambda_2 < 0, K = 0$

$$\Rightarrow (\sqrt{\lambda_1}x - \sqrt{-\lambda_2}y)(\sqrt{\lambda_1}x + \sqrt{-\lambda_2}y) = 0 \Rightarrow \text{Union of two lines.}$$

V: $\lambda_2 < 0, K < 0$

$$| \cdot (-1) \Rightarrow -\lambda_1 x^2 - \lambda_2 y^2 = K$$

VI: $\lambda_2 < 0, K > 0$

$$| : K \Rightarrow \frac{x^2}{\frac{K}{\lambda_1}} - \frac{y^2}{\frac{K}{-\lambda_2}} = 1 \Rightarrow \text{Hyperbola.}$$

VII: $\lambda_2 = 0$

$$\Rightarrow C: \lambda_1 x^2 + \lambda_2 y = K$$

VII.1) $\lambda_2 = 0, K > 0 \Rightarrow (\sqrt{\lambda_1}x - \sqrt{K})(\sqrt{\lambda_1}x + \sqrt{K}) = 0 \Rightarrow 2 \text{ lines.}$

VII.2) $\lambda_2 = 0, K = 0 \Rightarrow \text{double line } (x^2 = 0)$

VIII: $\lambda_2 = 0, K < 0$

$\Rightarrow \nexists$ real sol.

IX: $\lambda_2 \neq 0 \Rightarrow \frac{\lambda_1}{|\lambda_2|} x^2 = \frac{K}{|\lambda_2|} - \frac{\lambda_2}{|\lambda_2|} y$

$$\Rightarrow \dots \Rightarrow x^2 = \frac{|\lambda_2|}{\lambda_1} y. \Rightarrow \dots \Rightarrow \text{Parabola}$$

Parallel reflection in a hyperplane

Reflect

Project (v) point in $\mathbb{A}^2$ onto line $l: x+y-1=0$, in the direction of $(-2, -1)$.

$$l_p: \begin{cases} x = x_p - 2t \\ y = y_p - t \end{cases}$$

$P' = l \cap l_p =$ ~~proj.~~ reflection of P onto l s.t.

$l \cap l_p = H$ of $[PP']$.

$$\frac{P+P'}{2} = P_{H, v}(P) \Rightarrow P' = 2P_{H, v}(P) - P.$$

$$\begin{cases} l: x+y-1=0 \\ l_p: \begin{cases} x = x_p - 2t \\ y = y_p - t \end{cases} \end{cases}$$

$$[P']_k = \begin{bmatrix} -\frac{1}{3}x_p - \frac{4}{3}y_p + \frac{4}{3} \\ -\frac{2}{3}x_p + \frac{1}{3}y_p + \frac{2}{3} \end{bmatrix} = \underbrace{\frac{1}{3} \begin{bmatrix} -1 & -4 \\ -2 & 1 \end{bmatrix}}_A \begin{bmatrix} x_p \\ y_p \end{bmatrix} + \underbrace{\frac{2}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix}}_b$$

Generaliz: $H: a_1x_1 + \dots + a_nx_n + a_{n+1} = 0$ hyp.

$$l_p \cap H \Rightarrow a_1(p_1 + t \cdot v_1) + \dots + a_n(p_n + t \cdot v_n) + a_{n+1} = 0.$$

$$l \cap l_p: P' = P - 2 \cdot \frac{a^T P + a_{n+1}}{a^T v} \cdot v.$$

$$P' = P - 2 \cdot \frac{a^T P \cdot v}{a^T v} - 2 \cdot \frac{a_{n+1}}{a^T v} \cdot v$$

$$P' = P - 2 \cdot \frac{v \cdot a^T P}{a^T v} - 2 \cdot \frac{a_{n+1}}{v^T a} v$$

$$P' = P \left(I_n - 2 \cdot \frac{v \cdot a^T}{v^T a} \right) - 2 \cdot \frac{a_{n+1}}{v^T a} v$$

$$\hookrightarrow \text{ORTHONORMAL} \Rightarrow \left(I_n - 2 \cdot \frac{v \otimes a}{\langle v, a \rangle} \right)$$

(IF)

Homog. coordinates

$$\phi(x) = Ax + b$$

Homog. matrix: $\begin{bmatrix} A & b \\ 0 & 1 \end{bmatrix}$

$$\psi(x) = A'x + b'$$

$$\psi \circ \phi(x) = \psi(\phi(x)) = A'(Ax + b) + b' = (A'A)x + (A'b + b')$$

$$\hat{H}_{K',K}(\psi \circ \phi) = \begin{bmatrix} A'A & A'b + b' \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} A' & b' \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} A & b \\ 0 & 1 \end{bmatrix} = H_{\dots}(\psi) \cdot H_{\dots}(\phi)$$

composition of affine maps $\Rightarrow$ multiplication of homog. matrices.

Homog. coordinates:

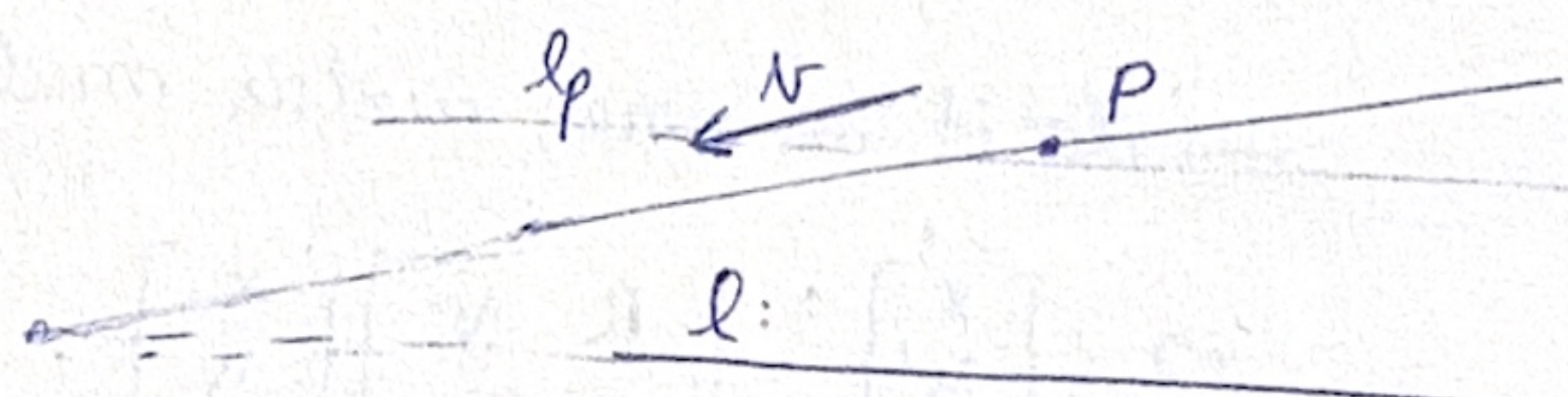
$$\begin{bmatrix} \phi(x) \\ 1 \end{bmatrix} = \begin{bmatrix} A & b \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x \\ 1 \end{bmatrix}$$

Parallel proj. on hyperplane

Project (4) point in $\mathbb{A}^2$ onto line $l: x+y-1=0$, in the direction of $v(-2,-1)$

$\Rightarrow$ l_p passing through P in dir. of v :

$$l_p: \begin{cases} x = x_p - 2t \\ y = y_p - t \end{cases}$$



$P' = l \cap l_p =$ projection onto l in dir. of v .

$l: y = -x + 1 \Rightarrow m_l = -1$

$l_p: \begin{cases} 2t = x_p - x \\ t = y_p - y \end{cases} \Rightarrow 2(y_p - y) = x_p - x$

$l_p =$ line in the direction of $v(-2,-1) \Rightarrow m_{l_p} = \frac{v_y}{v_x} = \frac{-1}{-2} = \frac{1}{2} \neq m_l$

$\Rightarrow l \cap l_p \neq \emptyset$

$$\begin{cases} l: x+y-1=0 \\ l_p: \begin{cases} x = x_p - 2t \\ y = y_p - t \end{cases} \end{cases} \Rightarrow \begin{cases} x_p - 2t + y_p - t - 1 = 0 \\ x_p + y_p - 3t - 1 = 0 \end{cases}$$

$$\Rightarrow x_p + y_p - 1 = 3t$$

$$\Rightarrow t = \frac{1}{3}(x_p + y_p - 1)$$

$$\Rightarrow x = x_p - 2t = x_p - \frac{2}{3}(x_p + y_p - 1)$$

$$\Rightarrow y = y_p - \frac{1}{3}(x_p + y_p - 1)$$

$$[P']_K = \begin{bmatrix} \frac{1}{3}x_p - \frac{2}{3}y_p + \frac{2}{3} \\ -\frac{1}{3}x_p + \frac{2}{3}y_p + \frac{1}{3} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x_p \\ y_p \end{bmatrix} + \frac{1}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Generaliz: $H: a_1x_1 + \dots + a_mx_m + a_{m+1} = 0$ hyp.

l_p passing through $P(p_1, \dots, p_m)$ and dir. vector $v(v_1, \dots, v_m)$ $l_p = \{P + tv : t \in \mathbb{R}\}$

$l_p \cap H \Rightarrow a_1(p_1 + t \cdot v_1) + \dots + a_m(p_m + t \cdot v_m) + a_{m+1} = 0$

$l \cap l_p: P' = P - \frac{a_1 p_1 + \dots + a_m p_m + a_{m+1}}{a_1 v_1 + \dots + a_m v_m} v = P - \frac{a^T \cdot P + a_{m+1}}{a^T \cdot v} \cdot v$

$$\begin{cases} p'_1 = p_1 + \mu v_1 \\ \vdots \\ p'_m = p_m + \mu v_m \end{cases}$$

$$= P - \frac{a^T \cdot P \cdot v}{a^T \cdot v} - \frac{a_{m+1}}{a^T \cdot v} \cdot v$$

$\hookrightarrow$ ORTHONORMAL $\Rightarrow \left(I_m - \frac{v \otimes a}{\langle v, a \rangle} \right)$

$$\begin{aligned} \langle a, v \rangle &= a^T v = \\ &= \langle v, a \rangle = v^T a \end{aligned}$$

Isometric invariants (D2)

2D Hyperquadric: $C: g_{11}x^2 + 2g_{12}xy + g_{22}y^2 + b_1x + b_2y + c = 0$

↓

$$C: g_{11}x^2 + 2g_{12}xy + g_{22}y^2 + 2b_1x + 2b_2y + c = 0$$

↓

$$\begin{bmatrix} x & y & 1 \end{bmatrix} \begin{bmatrix} g_{11} & g_{12} & b_1 \\ g_{21} & g_{22} & b_2 \\ b_1 & b_2 & c \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = 0$$

$\hat{Q}$ = symm. matrix =
= extended symm. matrix

→ rotation R followed by translation v : $X = RX' + v$

→ do it in one matrix multiplication

$$\Rightarrow \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} R & v \\ 0 & 1 \end{bmatrix}}_P \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\Downarrow$$

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = P^{-1} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\Downarrow$$

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} R^{-1} & -R^{-1}v \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix}$$

⇓

$$\begin{bmatrix} x' & y' & 1 \end{bmatrix} H^T \hat{Q} H \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = 0$$

$$\det(H^T \hat{Q} H) = \det(H^T) \cdot \det(\hat{Q}) \cdot \det(H) = \det(\hat{Q}) \leftarrow \text{INVARIANT}$$

INVARIANTS: $\begin{cases} \hat{D} = \det(\hat{Q}) \\ D = \det(Q) \\ T = \text{tr}(Q) \end{cases}$

→ 3x3 matrix (also constants)
→ 2x2 matrix (only quadratic terms: x^2, y^2, xy)

	$\hat{D} = \det(\hat{Q})$	$D = \det(Q)$	$T = \text{Tr}(Q)$
degenerate conic $\Leftarrow \hat{D} = 0$			point 2 lines / $\emptyset$ 2 lines
normal conic $\Leftarrow \hat{D} \neq 0$			

2)

11) $H = \begin{bmatrix} 6 & 2 \\ 2 & 9 \end{bmatrix} \rightarrow A, H \in SO(2)$ which diag. A?

$\hookrightarrow 6x^2 + 9y^2 + 4xy + 42 = 0.$

$\det(A - \lambda I_2) = \begin{vmatrix} 6-\lambda & 2 \\ 2 & 9-\lambda \end{vmatrix} \rightarrow \lambda_1 = 10 \rightarrow v_1 = (1, 2)$
 $\rightarrow \lambda_2 = 5 \rightarrow v_2 = (-2, 1).$

$|v_1| = \sqrt{5}; |v_2| = \sqrt{5}.$

$H_{ee'} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \rightarrow \det = \frac{1}{5} \cdot (-4 - 1) = -1.$

$\hookrightarrow 1 \cdot (-1)$

$\Rightarrow H_{ee'} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \rightarrow \det = 1 \checkmark$

$H \cdot H^T = \dots = I_2 \quad \left. \begin{matrix} \det = 1 \\ \end{matrix} \right\} \Rightarrow \in SO(2).$

3)

2)

12) Get into isometric can. eq: $x^2 - 4xy + y^2 - 6x + 2y + 1 = 0$

$Q = \begin{bmatrix} 1 & -2 \\ -2 & 1 \end{bmatrix}$

$\begin{vmatrix} 1-\lambda & -2 \\ -2 & 1-\lambda \end{vmatrix} \rightarrow \lambda_1 = -1 \rightarrow v_1 = (1, 1)$
 $\rightarrow \lambda_2 = 3 \rightarrow v_2 = (1, -1)$

$\Rightarrow |v_1| = \sqrt{2}, |v_2| = \sqrt{2}$

$\Rightarrow H_{BB'} = H_{ee'} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \rightarrow \det = \frac{1}{2} (-1 - 1) = -1 \text{ NOT!}$

$H_{BB'} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \Rightarrow \det = \frac{1}{2} (1 + 1) = 1 \checkmark.$

$\downarrow \downarrow$
 $v_2 \quad v_1$

Obs! $H_{BB'} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \Rightarrow \theta = -\frac{\pi}{4}.$

Eq. of quadratic curve:

$[x' \ y'] H^T Q H \begin{bmatrix} x' \\ y' \end{bmatrix} + [-6 \ 2] + 1 = 0$

$\Rightarrow 3 \underbrace{\left(x' - \frac{4}{3\sqrt{2}}\right)^2}_{x''} - \underbrace{\left(y' + \frac{2}{\sqrt{2}}\right)^2}_{y''} - \frac{1}{19} = 0 \Rightarrow$

Seminars - Redone

SEK5. 5.1. $\rightarrow$ proj. on Ox : $y=0 \Rightarrow 0 \cdot x + 1 \cdot y + 0 = 0$, $n = (0, 1)$.

$\rightarrow$ proj. on Oy : $x=0 \Rightarrow 1 \cdot x + 0 \cdot y + 0 = 0 \Rightarrow n = (1, 0)$.

• Better use proj, refl formula in a POINT.

• Once a point is proj. on H , proj. again does nothing.

• $\text{Ref} \circ \text{Ref} \Rightarrow \text{Id}$.

• $\text{Pr} \circ \text{Pr} \Rightarrow \text{Pr}$.

• Hyp. passing through origin: $H: \langle m, x \rangle = 0 \Leftrightarrow H: a^T x + a_{n+1} = 0$.
 $a_{n+1} = 0$

$a = m \rightarrow$ normal vector, length 1 $\Rightarrow |m| = 1$.

SEK6. 6.1. $\triangle ABC$ rotated by 90° around $C \rightarrow$ reflection in AB .

$$R_{C, 90^\circ}(X) = C + R_{90^\circ} \cdot (X - C) \dots = \begin{bmatrix} 5-y \\ 1+x \end{bmatrix}$$

$$\Rightarrow A' = \begin{bmatrix} 5-y_A \\ 1+x_A \end{bmatrix} = \dots$$

$$\Rightarrow B' = \dots$$

$$\Rightarrow C' = \dots$$

$$AB \div \frac{x - x_A}{x_B - x_A} = \frac{y - y_A}{y_B - y_A}$$

$$AB: \frac{x-1}{3} = \frac{y-1}{0} \Rightarrow y=1 \quad ??$$

6.2. Prove it's a rot.

$$a) \Phi \left(\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \right) = \frac{1}{5} \begin{bmatrix} 4 & -3 & 7 \\ 3 & 4 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \frac{1}{5} \begin{pmatrix} 4x - 3y + 7 \\ 3x + 4y - 1 \\ 1 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 4 & -3 \\ 3 & 4 \end{pmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \frac{1}{5} \begin{bmatrix} 7 \\ -1 \end{bmatrix}$$

$$\boxed{A^T A = I_2} \Rightarrow \text{rotation} \dots$$

$$\boxed{\det A = 1}$$

b)

$$2) \quad 3x^{u2} - y^{u2} = -\frac{1}{9} \quad | \cdot (-9)$$

$$\Rightarrow -27x^{u2} + 9y^{u2} = 1$$

$$\Rightarrow -\frac{x^{u2}}{\frac{1}{27}} + \frac{y^{u2}}{\frac{1}{9}} = 1$$

but $\frac{1}{27} < \frac{1}{9} \Rightarrow$ can form. ($a > b$)

$$x^{u2} = y^{u2}$$

$$y^{u2} = x^{u2}$$

$$\Rightarrow -\frac{y^{u2}}{\frac{1}{27}} + \frac{x^{u2}}{\frac{1}{9}} = 1 \quad \text{hyperbola.}$$

[3]. Same, but Lagrange method: $x^2 - 4xy + y^2 + 6x + 2y + 1 = 0$.

$$(2) \quad (x^2 - 4xy + 4y^2)^2 - 4y^2 + y^2 - 6x + 2y + 1 = 0$$

$$2) \quad \underbrace{(x - 2y)^2}_{x_1} \dots$$

$$y_1 = y$$

$$\boxed{\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} x - 2y \\ y \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}}$$

$$\Rightarrow x_1^2 - 3y_1^2 \dots$$

$$2) \quad \underbrace{(x_1 - 3)^2}_{x_2} + \underbrace{\left(\sqrt{3}y_1 + \frac{5}{\sqrt{3}}\right)^2}_{y_2} - \frac{1}{3} = 0$$

$$\Rightarrow \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 - 3 \\ \sqrt{3}y_1 + \frac{5}{\sqrt{3}} \end{bmatrix} = \begin{bmatrix} 1 & -3 \\ 0 & \sqrt{3} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} + \begin{bmatrix} -3 \\ \frac{5}{\sqrt{3}} \end{bmatrix}$$

$$\Rightarrow x_2^2 - y_2^2 = -\frac{1}{3} \quad | \cdot (-\frac{1}{3})$$

$$\Rightarrow \underbrace{\left(\frac{y_2}{\frac{1}{\sqrt{3}}}\right)^2}_{x_3} - \underbrace{\left(\frac{x_2}{\frac{1}{\sqrt{3}}}\right)^2}_{y_3} = 1 \Rightarrow \text{hyp.}$$

$$\begin{bmatrix} x_3 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 & \sqrt{3} \\ \sqrt{3} & 0 \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & \sqrt{3} \\ \sqrt{3} & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \sqrt{3} \end{bmatrix} \dots$$

↓
all changes of coords.